

Reversible Multibaker Transport: Primitive Orbits and the Corrected Twisted Zeta

HCS-C379 theorem and reproducibility package

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Abstract

We give a complete source-system theorem for the reversible multibaker on a finite ring, with a domain convention that resolves the binary endpoint ambiguity. On the invariant set of non-dyadic coordinates, every geometric periodic point is reconstructed from a mixed binary word. A primitive necklace of length d and displacement S lifts to $\gcd(L, S)$ cycles of least period $dL/\gcd(L, S)$, with exact winding, reversal and reciprocal stability multipliers. The finite tilted transport determinant has a Chebyshev formula. Its weighted primitive product contains two homogeneous symbolic cycles absent from the geometric domain; removing them gives an explicit all-order correction. We distinguish inverse-unstable weights from two-dimensional flat-trace weights.

Keywords: multibaker map; primitive orbit; winding cocycle; boundary coding; twisted determinant; deterministic diffusion.

中文摘要

本文在明确排除二进制边界的不变满测集上，建立有限环可逆多面包师映射的完整原始周期轨道分类。每个几何周期点由混合二进制词唯一重建；长度和位移决定提升后的最小周期、轨道条数、整数环绕数、反演配对及互为倒数的稳定乘子。有限维扭转输运行列式具有切比雪夫闭式。它的符号周期乘积含有两条不在几何定义域中的齐次边界轨道；精确去除这两项得到全阶校正公式，并明确区分不稳定乘子的逆权重与二维平坦迹权重。

关键词：多面包师映射；原始轨道；环绕余循环；边界编码；扭转行列式；确定性扩散。

1 The phase space and the endpoint problem

The classical multibaker is a deterministic area-preserving realization of nearest-neighbor diffusion; the model and its transport interpretation are established in the literature [1, 2]. The present contribution is an explicitly normalized proof and independent computational audit. We claim no priority for the classical diffusion mechanism. The point requiring particular care is that symbolic completeness and geometric completeness depend on the chosen boundary convention.

Let $\mathcal{D}$ denote the dyadic rationals in $[0, 1]$, let $X = \{x \in (0, 1) : x \notin \mathcal{D}\}$, and set

$$M_L = (\mathbb{Z}/L\mathbb{Z}) \times X \times X, \quad L \geq 1.$$

For $s = \lfloor 2x \rfloor$ define $d_s = 2s - 1$ and

$$B(j, x, y) = (j + d_s \bmod L, 2x - s, (y + s)/2). \quad (1)$$

Both cell length and time step equal one. Normalized cell counting measure times Lebesgue area gives an invariant probability measure on M_L . Removing dyadic coordinates removes a null set; it does affect the exact periodic-orbit convention, which is why it is stated before any census.

On a half-open unit square the all-one periodic code reconstructs the excluded point $(1, 1)$. Keeping that code while using a single-valued half-open baker therefore overcounts a geometric orbit. Our symmetric non-dyadic domain excludes both homogeneous codes and retains every mixed periodic code. This makes the reversal exact on the stated domain.

Lemma 1 (Coding and reversal). *The map B is bijectively conjugate to the two-sided binary shift with cell cocycle d_s , restricted to sequences whose two tails are not eventually constant. It preserves area on each branch. The involution $I(j, x, y) = (j, 1 - y, 1 - x)$ satisfies $IBI = B^{-1}$ and reverses the area form.*

Proof. Use the unique expansions $x = .s_0s_1\cdots$ and $y = .s_{-1}s_{-2}\cdots$. Formula (1) shifts the symbolic sequence and adds d_{s_0} to the cell. Non-dyadic expansions are unique and neither tail is eventually constant, so this coding and its inverse are defined without endpoint identifications. With $t = \lfloor 2y \rfloor$ one computes

$$B^{-1}(j, x, y) = (j - d_t, (x + t)/2, 2y - t).$$

Since $y \neq 1/2$, the first symbol of $1 - y$ is $1 - t$; substitution proves $IBI = B^{-1}$. The derivative on each branch is $\text{diag}(2, 1/2)$, while $I^*(dx \wedge dy) = -dx \wedge dy$. The images of the two branches partition the full-measure domain, establishing measure preservation as well as the local Jacobian statement. $\square$

2 All geometric periods, powers and winding

For a word $w = s_0 \cdots s_{n-1}$, write $a(w)$ for its binary integer, w^{rev} for its reversal, and $S(w) = \sum_t (2s_t - 1)$. A mixed word contains both symbols. A necklace is a word modulo cyclic rotation; it is primitive when its least word period equals its length.

Theorem 1 (Complete geometric atlas). *Every length- n mixed word satisfying $L \mid S(w)$ gives, for each cell j , exactly one point fixed by B^n , with*

$$x = \frac{a(w)}{2^n - 1}, \quad y = \frac{a(w^{\text{rev}})}{2^n - 1}. \quad (2)$$

These are all fixed points. A primitive mixed necklace of length d and displacement S lifts to exactly $g = \gcd(L, S)$ geometric cycles, each with

$$q = dL/g, \quad W = S/g. \quad (3)$$

Here q is the least geometric period, W is winding, and $\gcd(L, 0) = L$. Its r -fold repetition has time rq , winding rW , multipliers $2^{rq}, 2^{-rq}$ and phase $e^{ir\phi W}$.

Proof. Affine iteration gives $x_n = 2^n x - a(w)$ and $y_n = 2^{-n} y + 2^{-n} a(w^{\text{rev}})$. The fixed equations force (2). For mixed w , the resulting fractions lie strictly between zero and one and have odd reduced denominators greater than one. They are non-dyadic, and their periodic expansions realize the specified word. The cell closes precisely when $L \mid S(w)$. Conversely every periodic point has periodic coding by Lemma 1; the two homogeneous words would give corners outside the domain.

For a primitive necklace, the internal coordinates first return after d steps. At that time the cell translates by S . The order of this translation is $h = L/g$, so the first full return occurs at $q = dh$. There are Ld choices of cell and word phase; partition into cycles of length dh gives g distinct cycles. The total displacement is hS , yielding $W = hS/L = S/g$. Each derivative is diagonal, giving the stated powers and positive multipliers. No division by a reversal pairing has been made. $\square$

The involution sends a cycle to the complement of its reversed word, up to rotation. It sends S, W to $-S, -W$ and keeps q and multiplicity fixed. Self-reversing cycles are allowed. The distinction between word period and geometric period is substantial: the primitive word 001 has $d = 3$ and $S = -1$, so on an L -cell ring it gives one cycle of period $3L$, not L cycles of period three. The word 01 has $S = 0$, and gives L cycles of period two.

Corollary 1 (All-period fixed census). *For every $L, n \geq 1$,*

$$F_{L,n} = L \left(\sum_{\substack{0 \leq k \leq n \\ L \mid (2k-n)}} \binom{n}{k} - 2\mathbf{1}_{L \mid n} \right). \quad (4)$$

The number of geometric primitive cycles of period q is $q^{-1} \sum_{r \mid q} \mu(r) F_{L,q/r}$.

Proof. A length- n word with k ones has displacement $2k - n$ and there are $\binom{n}{k}$ such words. Each closing word permits all L cells. The two homogeneous words close precisely when $L \mid n$ and must be subtracted. Every period- n fixed point belongs to one unique primitive cycle whose period divides n ; Möbius inversion gives the last assertion. $\square$

3 The tilted transport determinant and geometric correction

For real ϕ , define the finite backward transport operator

$$(P_\phi f)_j = \frac{1}{2}e^{i\phi/L}f_{j+1} + \frac{1}{2}e^{-i\phi/L}f_{j-1}. \quad (5)$$

The two labelled transitions are added when their destination cells coincide. In particular, this definition does not discard either channel for $L = 1, 2$. We consider $D_L(z, \phi) = \det(I - zP_\phi)$. This is a finite source transport determinant with a specified observable space.

Theorem 2 (Determinant and complete trace). *The full spectrum, with algebraic multiplicities, is*

$$\lambda_k(\phi) = \cos \frac{2\pi k + \phi}{L}, \quad 0 \leq k < L.$$

Writing T_L for the Chebyshev polynomial,

$$D_L(z, \phi) = 2^{1-L}z^L(T_L(1/z) - \cos \phi) = \prod_{k=0}^{L-1} (1 - z\lambda_k(\phi)). \quad (6)$$

The displayed Laurent expression is understood after cancellation as a polynomial. For every $n \geq 1$,

$$\text{tr } P_\phi^n = L2^{-n} \sum_{\substack{w \in \{0,1\}^n \\ L|S(w)}} e^{i\phi S(w)/L}. \quad (7)$$

Proof. Fourier vectors $f_j = e^{2\pi i k j / L}$ diagonalize (5), giving the spectrum. For generic ϕ , these are distinct roots of $T_L(t) - \cos \phi$, since $T_L(\cos \theta) = \cos L\theta$. The leading coefficient is 2^{L-1} , including $L = 1$. Equality of the polynomials for generic ϕ extends by continuity to every real twist, including multiple roots. Substitution $t = 1/z$ proves (6). Expanding the matrix power as labelled steps proves (7): diagonal entries require cell closure, and the accumulated phase is $e^{i\phi S/L}$, not $e^{i\phi S}$. $\square$

Eigenvalues equal to zero do not produce finite determinant roots. Each nonzero eigenvalue contributes $z = 1/\lambda_k$ with its multiplicity, so the formula also specifies every degree drop and every source root count. Replacing ϕ by $\phi + 2\pi$ permutes the eigenvalue list.

Theorem 3 (Boundary-corrected geometric product). *Let p run over the geometric primitive cycles of Theorem 1. For real ϕ and $|z| < 1$, the absolutely convergent product is*

$$\prod_p \left(1 - (z/2)^{q_p} e^{i\phi W_p}\right)^{-1} = \frac{(1 - (z/2)^L e^{i\phi})(1 - (z/2)^L e^{-i\phi})}{D_L(z, \phi)}. \quad (8)$$

The quotient gives its rational continuation, retaining cancellations.

Proof. Since $|\text{tr } P_\phi^n| \leq L$, the series $\sum_{n \geq 1} z^n \text{tr } P_\phi^n / n$ converges absolutely for $|z| < 1$ and equals $-\log D_L$. Group each closed labelled walk by its unique primitive cycle and repetition. This yields the reciprocal Euler product over symbolic lattice cycles, with inverse-unstable weight 2^{-q} and phase $e^{i\phi W}$. Its only non-geometric cycles are the all-zero and all-one cycles. Each has least lattice period L , and their windings are $-1, +1$. Multiplying by the two corresponding factors removes them, proving (8). They remain distinct labelled cycles even for $L = 1, 2$. Alternatively the bound $F_{L,n} \leq L2^n$ controls the geometric logarithmic series directly. Both arguments justify regrouping without any cancellation being replaced by an absolute weight. $\square$

There is an additional weight boundary. The inverse unstable multiplier is 2^{-n} , but the two-dimensional fixed-point denominator is

$$|\det(I - DB^n)| = (2^n - 1)^2 / 2^n. \quad (9)$$

Its inverse is different. The finite transport determinant therefore has not been proved to be a Fredholm determinant of the full phase-space Perron operator. Neither an infinite-dimensional trace-class claim nor a target divisor is licensed by (6).

4 Conclusion and revision record

Round one adds the twisted transport determinant, its complete source spectrum and the geometric correction for the two omitted homogeneous cycles. It also separates the transport weight from the full flat-trace denominator, closing the principal determinant-convention ambiguity.

References

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